SAEs for Scientific Discovery: Status Report September 2025

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Abstract

This report summarizes progress on using sparse autoencoders (SAEs) to discover morphological traits in biological images. We implemented Matryoshka SAEs with ReLU activation and trained SAEs on DINOv3 activations for two datasets (FishVista and the Heliconius butterflies). We developed aspect-aware resizing for DINOv3 and comprehensive metrics logging for hyperparameter optimization. Next steps focus on Heliconius butterfly analysis for novel trait discovery, and systematic feature evaluations using FishVista and ADE20K's pixel-level semantic labels.

Introduction

Large vision transformers learn rich, structured representations of the visual world through pretraining on massive datasets. These models encode complex patterns—textures, shapes, object parts, and semantic relationships–within their high-dimensional activation spaces. However, this learned knowledge remains locked in dense, entangled representations that resist human interpretation. Sparse autoencoders (SAEs) offer a promising approach to extract these learned structures into discrete, interpretable features that we can examine through collections of maximally-activating images.

We aim to use SAEs to automatically discover and isolate meaningful morphological traits from vision transformer representations. SAEs decompose dense neural activations into sparse combinations of learned features, where each feature can be understood through the example images (anywhere from 5 to 20) that most strongly activate it. By training SAEs on biological vision models, we aim to create a computational lens for examining what these models have learned about organismal morphology, from wing patterns to body shapes to fine-grained anatomical structures.

The core challenge is demonstrating that SAE features correspond to biologically meaningful traits rather than arbitrary visual patterns. Without ground truth labels for most discoverable traits, evaluating features requires creative validation strategies that combine quantitative metrics with expert biological assessment. This challenge becomes an opportunity when studying mimicry systems, where novel trait discoveries could reveal previously unrecognized convergent features.

Figure 1: **Top:** A visual explanation of a known diagnostic trait (source) for distinguishing *H. erato notabilis* (left) and *H. melpomene plesseni* (right). **Bottom**: Top two images for feature DINOv3-16K/2735, which highlight the same parts of the wing that identifies *notabilis*.

We train SAEs on BioCLIP 2 and DINOv3 across three datasets: ADE20K for general visual under-

Date	Goal
September 22, 2025	Background-filtered visual examples of high-resolution (1920 patches/image) SAE features on DINOv3 with Heliconius butterflies.
September 29, 2025	Automatic *quantitative* evaluation of FishVista feature quality for SAEs from different layers of DINOv3.
October 6, 2025	
November 10, 2025	*(Probable CVPR 2026 submission deadline)*

Table 1: Approximate timeline for every Monday for the rest of 2025.

standing, FishVista for aquatic morphology, and Heliconius butterflies for studying mimicry. We automatically evaluate feature quality using segmentation labels from ADE20K and FishVista, establishing baseline performance against PCA, k-means, and NMF decomposition methods. For Heliconius, we aim to discover novel diagnostic traits between mimetic pairs (*H. melpomene* and *H. erato* subspecies), potentially revealing features that have escaped human observation but are captured in the models' learned representations.

Progress

- Jake implemented Matryoshka SAE[1] objective and functions. This leads to qualitatively better features on FishVista.
- We trained and visualized Matryoshka SAEs on DINOv3 activations on FishVista and butterflies.
- Found one qualitative example of a diagnostic trait in Heliconius mimics between *H. erato notabilis* and *H. melpomene plesseni*. See Figure 1 for the diagnostic feature.
- Developed a supervised linear probe for FishVista semantic segmentation. This should be an upper bound on the mean quality of an SAE feature.
- Added comprehensive metrics logging (MSE, explained variance, L0/L1 sparsity, dead unit percentage) to SAE training.
- Developed aspect-aware image resizing for DINOv3.

[1] Learning Multi-Level Features with Matryoshka Sparse Autoencoders

Next Steps & Timeline

We plan to submit to Nature Machine Intelligence. In order to do so, I think we must:

1. Describe a new scientific discovery (a novel and meaningful trait, ideally 3-5).
2. Compare against baseline methods on a variety of datasets, using statistical tests to validate improvements over multiple metrics.
3. (Ideally) develop an update to SAEs in order to enable this

See Table 1 for a plan on achieving this goal.